Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1048 to the nearest hundred.

2. Round 10.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Find |0| and describe it as a distance.

6. Find the opposite of 9.

7. Reflect (-3, 4) across the y-axis.

8. A triangle has base 9 cm and height 4 cm. Find its area.

9. A rectangle is 7 cm by 3 cm. Find its area.

10. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

11. Find the surface area of a closed 3 by 3 by 4 cm box.

12. A box has edges $\frac{1}{2}$, $\frac{2}{3}$, and 2 meters. Find the volume.

13. Write an inequality for x that means at least 9. Is the boundary included?

14. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

15. Start at −3. Move 5 units right. Where do you land?

16. Solve 2x + 3 = 11.

17. Write signed numbers for 8 meters below sea level and 3 meters above it. State what zero represents.

18. Order these values from least to greatest: $\frac{3}{2}$, -8, 0, $\frac{-3}{2}$.

CHALLENGE / Explain your plan and check the result

C1. A safe freezer temperature t must be at least −8°C but below −3°C. Test −9, −8, −5, −3, and 0. Graph the allowed interval and explain why −8 is included while −3 is not.

C2. A bird starts at (−2, 4), flies straight down 7 units, then straight right 6 units. Give the two new positions and total distance. Another bird flies right 6 then down 7 from the same start. Does it finish at the same point? Explain.